

Workings from Bartle-Sherbet and Lebl

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21st January 2026

Lebl

Proof of Strong Induction

Let $P(n)$ be statement depending on a natural number n . Suppose that

- $P(1)$ is true
- If $P(k)$ is true for all $1, 2, \dots, k$ then $P(k + 1)$ is true

Let S be the set of natural numbers s such that $P(s)$ is not true. Suppose for contradiction that S is non-empty, then S has a least element, say m , by the well ordering property. We know $1 \notin S$ since $P(1)$ is true. Then $m > 1$ and $m - 1$ is a natural number as well. Then $P(m - 1)$ must be true since m , is the least element for which $P(s)$ is not true, this also is true for all $s < m$, but then $P(m - 1 + 1)$ must be true. A contradiction, thus m is not in S , meaning there is not a least element in S , i.e. it is empty. Thus $P(k)$ is true for all natural numbers

□¹

¹meh

Bartle-Sherbet

Exercise 1.1.7

Let $A := B := x \in \mathbb{R} : -1 \leq x \leq 1$ and consider the subset $C := (x, y) : x^2 + y^2 = 1$ of $A \times B$. Is this set a function? Explain.

If C , a subset of $A \times B$, is a function from $A \rightarrow B$, for each $a \in A$, there exists a unique $b \in B$ with $(a, b) \in C$. For each $x \in A$ the function which gives the $y \in B$ is $y = \pm\sqrt{1-x^2}$, for this y to not be in B , we either need $1-x^2 < 0$ or $|\sqrt{1-x^2}| > 1$.

The restrictions these put on x are $-1 < x < 1$ which are all in the set A .

But then a problem arises with the uniqueness of the y , when $x \neq \pm 1$, the function has two possible values, say $x = 0.5$ then $y = \pm\frac{\sqrt{3}}{2}$. A contradiction to our definition.

Thus C is not a function.